Tech support scam pop-ups

Article KB0010085 | Owner: Security Operations | Subcategory: Antivirus / malware alert | Applies to: Windows 11 laptops (Dell Latitude 5440/7440, Lenovo ThinkPad T14/X1 Carbon, HP EliteBook 840) managed by Microsoft Intune, macOS devices running macOS 14/15, Any Acme Global device accessing the internet via corporate Wi-Fi, VPN, or guest networks

Summary

Tech support scam pop-ups are fraudulent browser windows that mimic antivirus alerts, urging users to call a fake support number or download malicious software. They are usually harmless if ignored, but can indicate exposure to risky sites. This article helps users recognise the scam, safely close the pop-up, run a Defender scan, and know when to involve Security Operations.

Symptoms

- Full-screen or modal browser pop-up titled "Windows Defender Alert" or "Microsoft Security Essentials"
- Message stating "Your computer is infected with viruses and spyware. Call support now at 1-800-XXX-XXXX"
- Fake scan progress bar showing dozens of threats detected
- Audio voice-over urging immediate action
- Inability to close the window via the usual X button; clicking anywhere may trigger a download prompt

Common causes

- Visiting compromised or malicious websites that serve malvertising ads
- Clicking on deceptive links in phishing emails or social media posts
- Outdated browser extensions or plugins that inject unwanted scripts
- Redirects from legitimate sites via hijacked ad networks

Initial checks

Before taking any action, confirm that the alert is indeed a browser-based pop-up and not a legitimate Windows Defender notification. Legitimate Defender alerts appear inside the Windows Security app, not inside a browser window, and do not include a phone number to call.

User self-help procedure

Follow these steps to safely dismiss the scam pop-up, verify that no real malware is present, and restore normal browser behaviour. The procedure works on Windows laptops managed by Intune and on macOS devices using Safari, Chrome, or Edge.

1. Do not call the phone number shown in the pop-up and do not download any offered software.
2. If possible, close the browser tab or window by right-clicking the browser icon on the taskbar and selecting "Close window". If the pop-up blocks normal controls, press Ctrl?+?Shift?+?Esc to open Task Manager, locate the browser process (chrome.exe, msedge.exe, or safari.exe), select it, and click "End task".
3. Re-open your browser and navigate to a trusted site (e.g., https://it.acme.example) to confirm the browser works normally.
4. Open the Microsoft Defender Security Center: click the Start menu, type "Windows Security", and open the app. Choose "Virus & threat protection" -> "Quick scan". Allow the scan to complete.
5. If the quick scan reports no threats, run a full scan for extra assurance: under the same menu select "Scan options" -> "Full scan" -> "Scan now".
6. After the scan finishes, review the results. If Defender reports any quarantined items, note the names and proceed to the escalation section below. If the scan is clean, the pop-up was a scam and no further action is required.
7. Optionally, clear the browser cache and reset settings to remove any lingering scripts: in Chrome/Edge go to Settings -> Privacy and security -> Clear browsing data -> choose "Cached images and files" and "Cookies and other site data", then restart the browser.

Service-desk agent verification

When a user reports a tech-support scam pop-up, the agent must first determine whether the alert is purely a browser-based scam or if Defender has detected genuine malware. This section outlines the checks agents should

perform and the information to collect.

1. Ask the user to describe exactly what they saw, including any wording, phone numbers, and whether audio was played. Request a screenshot or a photo of the pop-up if possible.
2. Verify the device hostname and IP address via the Intune portal or by asking the user to run "ipconfig" in Command Prompt.
3. Instruct the user to open Windows Security and run a Quick scan. Record the scan result (clean, threats found, or quarantined items).
4. If the quick scan shows threats, have the user run a Full scan and capture the final report.
5. Collect the Defender scan logs (export from Windows Security -> Protection history) and any remediation steps already taken (e.g., browser closed, cache cleared).

Verification and closure

After the user completes the self-help steps or after the agent runs scans, confirm that the system is clean and the pop-up does not return. Document the outcome and close the ticket, or escalate if malware is found.

1. Ask the user to reboot the device and browse to a few regular sites to ensure no pop-up reappears.
2. If Defender reported threats and they were quarantined, confirm quarantine status in Windows Security -> Protection history.
3. If the system remains clean, close the ticket with resolution: "Tech support scam pop-up dismissed; Defender scan clean; no action required."
4. If Defender found active malware that cannot be quarantined, proceed to escalation.

When to escalate

Escalate to the Security Operations team when Defender detects genuine malware that cannot be quarantined, the pop-up persists after browser reset and Task Manager closure, or the user indicates they have called the fake support number or downloaded software. Include the following details in the escalation ticket: device name, serial number, Intune device ID, timestamp of the pop-up, exact text or screenshot of the alert, Defender scan logs (Protection history export), and a summary of any user-performed remediation steps.

Keywords: tech support scam, fake virus alert, browser pop-up, Defender alert, malvertising, phone number scam, Windows Security, quick scan, full scan, Security Operations